Problem Set 1 - IC104

1 Solutions

Essential Mathematics for Economic Analysis, 4th edition

Page 9. Question 5
Answers:

$$a)\frac{p^{24}p^3}{p^4p} = \frac{p^{27}}{p^5} = p^{27-5} = p^{22}$$
$$b)\frac{a^4b^{-3}}{(a^2b^{-3})^2} = \frac{a^4b^{-3}}{a^4b^{-6}} = a^{4-4}b^{-3-(-6)} = b^3$$
$$c)\frac{3^4(3^2)^6}{(-3)^{15}3^7} = \frac{3^43^{12}}{(-1)^{15}3^{15}3^7} = \frac{3^{16}}{(-1)3^{22}} = -3^{-6}$$
$$d)\frac{p^\gamma(pq)^\sigma}{p^{2\gamma+\sigma}q^{\sigma-2}} = \frac{p^{\gamma+\sigma}q^\sigma}{p^{2\gamma+\sigma}q^{\sigma-2}} = \frac{p^{-\gamma}}{q^{-2}} = p^{-\gamma}q^2$$

Page 14. Question 6
Answers:

$$a)(x+2y)^2 = x^2 + 2\cdot x\cdot 2y + (2y)^2 = x^2 + 4xy + 4y^2$$
$$b)(\frac{1}{x} - x)^2 = (\frac{1}{x})^2 - 2\frac{1}{x}x + x^2 = \frac{1}{x^2} - 2 + x^2$$
$$c)(3u-5v)^2 = 9u^2 - 30uv + 25v^2\, d)$$
$$(2\mathrm{x} - 5w)(2\mathrm{x} + 5w) = 4\mathrm{x}^2 - 25w^2$$

Page 18. Question 2 a) b) c) d) e) f): Simplify the various expressions:

Answers:

$$a)\frac{x}{10}-\frac{3x}{10}+\frac{17x}{10}=\frac{x-3x+17x}{10}=\frac{15x}{10}=\frac{3x}{2}$$

$$b)\frac{9a}{10}-\frac{a}{2}+\frac{a}{5}=\frac{9a-5(a)+2(a)}{10}=\frac{6a}{10}=\frac{3a}{5}$$

$$c)\frac{b+2}{10}-\frac{3b}{15}+\frac{b}{10}=\frac{3(b+2)-2(3b)+3(b)}{30}=\frac{3b+6-6b+3b}{30}=\frac{6}{30}=\frac{1}{5}$$

$$d)\frac{x+2}{3}+\frac{1-3x}{4}=\frac{4(x+2)+3(1-3x)}{12}=\frac{4x+8+3-9x}{12}=\frac{-5x+11}{12}$$

$$e)\frac{3}{2b}-\frac{5}{3b}=\frac{3(3)-2(5)}{6b}=\frac{9-10}{6b}=\frac{-1}{6b}$$

$$f)\frac{3a-2}{3a}-\frac{2b-1}{2b}+\frac{4b+3a}{6ab}=\frac{2b(3a-2)-3a(2b-1)+(4b+3a)}{6ab}$$

$$=\frac{6ab-4b-6ab+3a+4b+3a}{6ab}=\frac{6a}{6ab}=\frac{1}{b}$$

Page 22. Question 3 a) c) e)

$$a)\sqrt{x}=9$$
$$x=9^2=81$$

$$c)\sqrt{x}\cdot\sqrt{4}=4$$
$$\sqrt{4x}=4$$
$$4x=16$$
$$x=4$$

$$e)\sqrt{(x+2)}=25$$
$$x+2=25^2\Rightarrow x=623$$

Page 45. Question 1 a) c) e)

$a) x_1 = 0, \quad x_2 = 15$

$c) q_1 = 3, \quad q_2 = -4$

$e) x^2 + x = 2x^2 - 2x \Rightarrow x^2 - 3x = 0 \Rightarrow x(x-3) = 0 \Rightarrow x_1 = 0, \quad x_2 = 3$

Page 47. Question 2 a) b) c)

$$
\begin{array}{rll}
a) & 5x + 2y = 3 & (i) \\
 & 2x + 3y = -1 & (ii) \\
 & & \\
(i) * 3 & 15x + 6y = 9 & (iii) \\
(ii) * 2 & 4x + 6y = -2 & (iv) \\
(iii) - (iv) & 11x = 11 & \\
 & x = 1 & \\
sub.in(i) & 5 + 2y = 3 & \\
 & 2y = -2 & \\
 & y = -1 & \\
 & x = 1, \quad y = -1 &
\end{array}
$$

$$
\begin{array}{rll}
b) & x - 3y = -25 & (i) \\
 & 4x + 5y = 19 & (ii) \\
 & & \\
(i) * 4 & 4x - 12y = -100 & (iii) \\
(ii) - (iii) & 17y = 119 & \\
 & y = 7 & \\
sub.in(i) & x - 21 = -25 & \\
 & x = -4 &
\end{array}
$$

$$
\begin{array}{rll}
c) & 2x + 3y = 3 & (i) \\
 & 6x + 6y = -1 & (ii) \\
(i) * 3 & 6x + 9y = 9 & (iii) \\
(iii) - (ii) & 3y = 10 & \\
 & y = \frac{10}{3} & \\
sub.in(i) & 2x + 10 = 3 & \\
 & 2x = -10 & \\
 & x = -\frac{7}{2} &
\end{array}
$$

Page 54. Question 3 a) b) e)

$$a) \sum_{i=1}^{n} 4i$$
$$b) \sum_{i=1}^{n} i^3$$
$$e) \sum_{n=1}^{5} 3^n x^n$$

Page 98. Question 2
Equilibrium occurs where D = S:

$$a) 75 - 3p = 20 + 2p \Rightarrow p = 11$$
$$b) 100 - 0.5p = 10 + 0.5p \Rightarrow p = 90$$